TEXORA · LECTURE SHEET

Design and Constructions of Basic Profiles

Designers do not order “a fancy yarn” — they order a profile: loops of a stated size at a stated spacing, or slubs of a stated length and thickness. This sheet describes the construction and appearance of the eight basic profiles and the three design parameters — effect length, spacing and frequency — that specify any of them.

YM-4103 · SPECIAL YARN MANUFACTURING · TOPIC 3 · LEVEL 4 TERM 1 · YARN MANUFACTURING SPECIALIZATION

1. What is a profile?

The **profile** of a fancy yarn is the shape its effect takes along the yarn length: what one repeat of the effect looks like in longitudinal view, how large it is, and how it is joined to the plain sections between effects. Every commercial fancy yarn is one basic profile, a combination of two profiles, or a profile varied by colour. Learning the eight basic profiles below — loop, snarl, slub, spiral, knot, chenille, gimp and stripe — gives the vocabulary for specifying, producing and inspecting all of them.

2. The eight basic profiles

2.1 Loop yarn (boucle)

Construction: a soft, low-twist effect yarn is overfed (commonly well above 20%, rising with loop size) onto a core yarn and twisted with it; a binder yarn applied in a second, opposite twisting locks the loops open. The surplus effect length buckles into closed, rounded loops standing off the core on all sides. **Appearance:** a curly, three-dimensional surface — the classic boucle jacketing and coating texture. Loop size is controlled by overfeed and effect-yarn stiffness; loop security is controlled entirely by the binder. A variant with elongated, wavy loops rather than closed circles is sold as wave or mohair-look loop yarn.

2.2 Snarl yarn (fringe yarn)

Construction: the effect yarn is a highly twisted, torque-lively yarn overfed onto a core. Released from tension, a lively yarn doubles back on itself; trapped by twisting and a binder, those doublings become tight, protruding snarls — small twisted pigtails sticking out of the yarn. **Appearance:** a spiky, fringed surface with strong texture and movement. Snarl formation therefore needs the opposite effect-yarn

character from loop yarn: high twist and liveliness, not softness. Insufficient binder lets snarls pull out; excessive core tension straightens them before they can form.

2.3 Slub / thick-place yarn

Construction: the yarn is a single spun yarn whose thickness is varied along its length by programmed drafting — periodically the drafting rollers slow or the delivery arrangement changes so that a short length receives less draft and emerges thicker (a slub), returning to the fine base count between slubs. No separate core or binder is needed; the slub is part of the yarn itself, which is why slub yarn is strong enough for warp use in moderation. **Appearance:** elongated thick places with tapered ends, giving fabrics a hand-spun, linen-like irregular stripe. Modern ring frames store slub patterns (length, thickness, spacing, random variation) as programmes; fully treated in Topic 4.

2.4 Spiral / corkscrew yarn

Construction: two yarns of different thickness, colour or character are twisted together with one fed under higher tension or at a differential rate, so the straighter, tenser component forms the axis and the slacker component winds around it in a regular open helix. No separate binder is required — the two components bind each other. **Appearance:** a regular barber-pole helix, striking when the two components contrast in colour. Spiral regularity depends on holding the tension difference constant; tension drift turns a spiral into an uneven wrap.

2.5 Knot / knop yarn

Construction: the effect yarn is wrapped repeatedly around the core at one stationary point — delivery of the assembly is effectively paused or the wrapping is concentrated — building a compact knob of wound yarn, after which normal running resumes and a plain interval follows before the next knot. Hollow-spindle and fancy twisting machines produce knots by programmed intermittent feed. **Appearance:** distinct beads or knobs spaced along an otherwise plain yarn, giving spotted, point-effect fabrics (knop yarns are a staple of dress tweeds). Knot size = number of wraps at the point; knot security depends on a binder or on tight wrapping.

2.6 Chenille yarn

Construction: fundamentally different — a pile is manufactured, not an overfeed effect. On the chenille machine, pile yarns are wrapped around a pair of spaced gauge blades; two core yarns are laid either side and twisted/pressed to grip the wraps; a rotary knife cutting along the centre of the blades slices every wrap, and the cut ends spring out radially, held by the core yarns. One wrapping yields two chenille yarns. **Appearance:** a soft velvet cord, round in cross-section, like a caterpillar — hence its name (French for caterpillar). Pile density and length (set by blade gauge) control fullness; the weakness of chenille is pile loss if the core grip is poor, so core construction and any adhesive/backing treatment are critical quality points.

2.7 Gimp yarn

Construction: a fine yarn (or filament) is wrapped closely and evenly around a thicker core yarn or cord, completely covering it with a tight helical sheath, often then wrapped again in the opposite direction. The core gives bulk and the wrapping gives a fine, wavy, corded surface. **Appearance:** a firm, rounded, slightly wavy cord with a fine ribbed texture, used in trimmings, furnishing and decorative weaving. Gimp is essentially a perfectly regular, fully-covering wrap — the disciplined opposite of the open spiral.

2.8 Stripe yarn (colour-effect / intermittent yarn)

Construction: the yarn itself stays structurally plain, but its colour or material changes along its length in programmed blocks — by exchanging effect yarns, by intermittent dyeing/printing of the yarn (space dyeing), or by feeding differently coloured rovings in sequence. **Appearance:** longitudinal colour stripes in the yarn that become broken colour patterns (checks, flecks, ikat-like effects) in fabric. The design variable is block length matched to fabric repeat: a stripe block that is an exact multiple of the pick length gives columns; a deliberately mismatched block gives scattered colour.

PROFILE	EFFECT MECHANISM	BINDER NEEDED?	SIGNATURE APPEARANCE
Loop	Soft effect yarn overfed, buckles to loops	Yes — locks loops open	Curly closed loops (boucle)
Snarl	Lively high-twist yarn doubles back	Yes	Tight protruding pigtails
Slub	Programmed drafting variation in one yarn	No — integral to yarn	Tapered thick places
Spiral	Two yarns, differential tension	No — self-binding	Regular open helix
Knot / knop	Wraps concentrated at a point	Usually	Spaced beads on plain yarn
Chenille	Wrapped pile cut between blades	Core grip instead	Velvet caterpillar cord
Gimp	Fine yarn fully wraps thick core	Wrap itself locks	Firm ribbed cord
Stripe	Colour/material changed in blocks	As base yarn	Longitudinal colour blocks

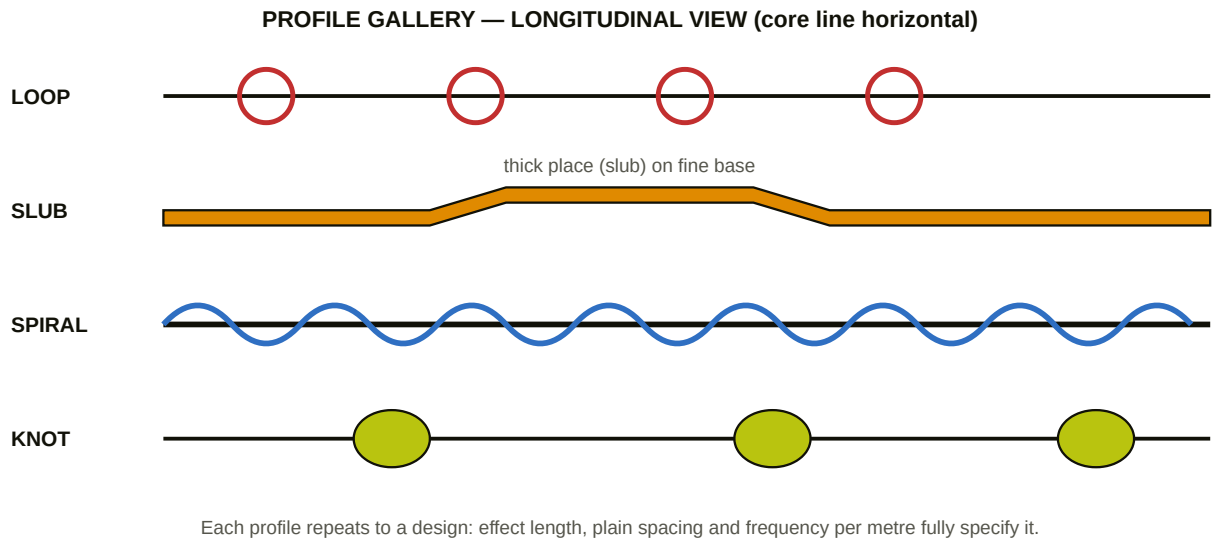


Figure 1 — Profile gallery. Loop, slub, spiral and knot yarns in longitudinal view — the four profiles most often combined in commercial fancy yarns.

3. Profile design parameters

Any profile, however complex its appearance, is specified for production and inspection by three length parameters plus the size of the effect itself:

- **Effect length** — the delivered length occupied by one effect: the length of one slub, the length over which a knot is built, the length of one colour block. For loop and spiral yarns the effect is continuous, and the corresponding parameter is loop size / helix pitch instead.
- **Spacing (plain length)** — the delivered length of plain yarn between successive effects. Spacing may be constant (regular design) or programmed to vary within stated limits (random / pseudo-random design, preferred where a mechanical repeat would show as a fabric pattern fault).
- **Frequency** — number of effects per unit delivered length, the reciprocal of the repeat length:

$$\text{Repeat length} = \text{effect length} + \text{spacing} \quad \cdot \quad \text{Frequency} = 1 / \text{repeat length}$$

WORKED EXAMPLE — FREQUENCY FROM A DESIGN REPEAT

A knop design has an effect (knot build) length of 1 cm and a plain spacing of 24 cm.
Repeat length = 1 + 24 = 25 cm = 0.25 m.
Frequency = 1 / 0.25 = **4 knots per metre** (400 knots per 100 m of yarn).
Design check: in a weft pattern using 2,500 picks of 1.4 m, total weft = 3,500 m, carrying about 3,500 × 4 = **14,000 knots** — the figure the buyer actually sees as spot density in the cloth.

Effect size completes the specification: slub thickness as a multiple of base count (e.g. slub = 2 × base tex over its length), loop diameter, knot diameter, chenille pile length. A complete design sheet states

profile name, component yarns and counts, overfeed ratio, effect length, spacing (or its variation rule), frequency, effect size, and colour order. Anything less leaves the effect to the machine operator's judgement — and two batches will not match.

4. Combining profiles and quality inspection

Commercial yarns routinely combine profiles: a slub yarn with a colour stripe, a loop yarn built on a spiral core, knots added to a gimp. Combination multiplies design freedom but also multiplies the inspection task. Inspection of fancy yarns is primarily **visual and length-based**: wind a measured length on a black board (or use a profile projector / camera system), and check effect size, spacing and frequency against the design sheet, plus binder security by a gentle pull test on effects. Count and strength are tested as for ordinary yarns, but a fancy yarn that passes every laboratory test and fails the board inspection is a reject — its value is its profile.

Glossary

Profile — longitudinal shape and arrangement of the effect in a fancy yarn. **Boucle** — loop-yarn fabric effect; the loop yarn itself. **Snarl** — tight self-doubled pigtail of a lively yarn. **Slub** — deliberate tapered thick place from varied drafting. **Knop** — bead of yarn wrapped at one point (knot yarn). **Chenille** — cut-pile velvet-cord yarn. **Gimp** — thick core fully covered by a fine tight wrap. **Effect length** — delivered length of one effect. **Repeat length** — effect length plus spacing. **Frequency** — effects per unit length, 1 / repeat length.

Key points

• Eight basic profiles: loop, snarl, slub, spiral, knot, chenille, gimp, stripe — every commercial fancy yarn is one or a combination.
• Loop needs a soft effect yarn; snarl needs a lively high-twist one — opposite feedstock characters for superficially similar textures.
• Slub is integral to a single yarn (no core/binder); chenille is a cut pile, not an overfeed effect.
• Any profile is fully specified by effect length, spacing and frequency (frequency = 1 / repeat length) plus effect size.
• Inspection is by board winding against the design sheet — profile conformity is the primary quality criterion.

Practice

1. A slub design has slub length 6 cm and spacing 44 cm. Calculate the repeat length and the number of slubs per 100 m.

2. A fabric shows tight spiky pigtails pulling out of the yarn surface. Name the profile, state the effect-yarn character that creates it, and name the component whose failure lets the pigtails pull out.
3. Distinguish gimp from spiral under construction, coverage and appearance.
4. Write a complete design-sheet specification (all parameters from Section 3) for a dress-fabric knot yarn of your own design, using plausible values.

Answer key

1. Repeat = 6 + 44 = 50 cm = 0.50 m. Frequency = 1 / 0.50 = **2 slubs per metre** = **200 slubs per 100 m**.
2. **Snarl** yarn. Created by a **high-twist, torque-lively** effect yarn doubling back on itself under overfeed. Pull-out is a **binder** failure — insufficient wraps or binder tension to lock the snarls.
3. Gimp: a fine yarn wraps a thick core closely until the core is *fully covered*; surface is a firm, fine-ribbed cord. Spiral: two comparable yarns twist together with differential tension, one winding in an *open* helix around the other, which remains visible; surface is a bold barber-pole helix, especially in contrasting colours.
4. Example of a complete answer: Profile — knot/knop. Core: 20 tex cotton/polyester spun yarn. Effect: 30 tex contrasting viscose spun yarn. Binder: 8 tex polyester filament. Knot build: effect wrapped to approx. 5 mm diameter over 1 cm effect length. Spacing: 24 cm plain, varied ±20% pseudo-randomly. Frequency: 4 per metre nominal (repeat 25 cm). Colour order: single contrast colour throughout. Any specification covering profile, components, effect length, spacing rule, frequency and effect size earns full credit.
